

E-COMMERCE DATABASE SYSTEM

Requirement Analysis

1. Project Overview

This project is a simple E-Commerce Database System. It stores details about customers, products, sellers, orders and payments. It helps to manage online shopping information easily.

2. E-Commerce Business Process

- Customer registers in the shopping application.
- Customer views products.
- Customer selects a product and places an order.
- Payment is made for the order.
- Seller receives the order and processes it.
- The product is delivered to the customer.

3. Customer Management Requirements

- Add customer details.
- View customer details.
- Update customer details.
- Delete customer details.
- Store customer orders and payments.

4. Major Entities

The five main entities are:

- Customer
- Product
- Seller
- Order
- Payment

5. Entities and Attributes

Entity	Attributes
Customer	Customer_ID, Name, Email, Phone, Address
Seller	Seller_ID, Seller_Name, Email, Phone
Product	Product_ID, Product_Name, Price, Stock, Seller_ID
Order	Order_ID, Customer_ID, Product_ID, Quantity, Order_Date, Status

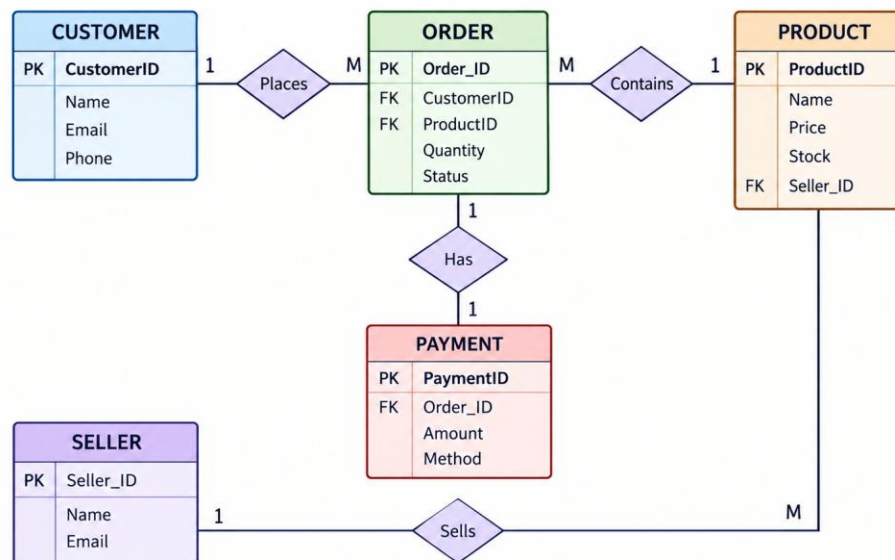
Payment	Payment_ID, Order_ID, Payment_Date, Amount, Payment_Method, Payment_Status
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6. Relationships Between Entities

- Customer → Order: One customer can place many orders.
- Seller → Product: One seller can sell many products.
- Product → Order: A product can be ordered by customers.
- Order → Payment: An order can have a payment.

7. ER Diagram

ER Diagram - E-Commerce Database



8. Functional Requirements

- Customer registration and management.
- Product management.
- Seller management.
- Order management.
- Payment management.
- Search and display records.
- Update and delete records.
- Generate simple database views.

9. Review Questions and Answers

Q. What is a primary key?

Answer: It uniquely identifies each record in a table.

Q. What is a foreign key?

Answer: It connects two tables.

Q. What is NOT NULL?

Answer: It makes a column compulsory.

Q. What is UNIQUE?

Answer: It prevents duplicate values.

Q. What is CRUD?

Answer: Create, Read, Update and Delete.

Q. What is a view?

Answer: A view is a virtual table based on an SQL query.

10. SQL Script

Create Database

```
DROP DATABASE IF EXISTS ecommerce;
```

```
CREATE DATABASE ecommerce;
```

```
USE ecommerce;
```

Create Customer Table

```
CREATE TABLE Customer (  
    Customer_ID INT PRIMARY KEY,  
    Name VARCHAR(50) NOT NULL,  
    Email VARCHAR(100) NOT NULL UNIQUE,  
    Phone VARCHAR(15) UNIQUE,  
    Address VARCHAR(100) NOT NULL  
);
```

Create Seller Table

```
CREATE TABLE Seller (  
    Seller_ID INT PRIMARY KEY,
```

```
Seller_Name VARCHAR(50) NOT NULL,  
Email VARCHAR(100) NOT NULL UNIQUE,  
Phone VARCHAR(15) UNIQUE  
);
```

Create Product Table

```
CREATE TABLE Product (  
    Product_ID INT PRIMARY KEY,  
    Product_Name VARCHAR(100) NOT NULL,  
    Price DECIMAL(10,2) NOT NULL,  
    Stock INT NOT NULL,  
    Seller_ID INT,  
    FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID)  
);
```

Create Orders Table

```
CREATE TABLE Orders (  
    Order_ID INT PRIMARY KEY,  
    Customer_ID INT,  
    Product_ID INT,  
    Quantity INT NOT NULL,  
    Order_Date DATE NOT NULL,  
    Status VARCHAR(20) NOT NULL,  
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),  
    FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
);
```

Create Payment Table

```
CREATE TABLE Payment (  
    Payment_ID INT PRIMARY KEY,  
    Order_ID INT,
```

```
Payment_Date DATE NOT NULL,  
Amount DECIMAL(10,2) NOT NULL,  
Payment_Method VARCHAR(30) NOT NULL,  
Payment_Status VARCHAR(20) NOT NULL,  
FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID)  
);
```

Insert Customer Data

```
INSERT INTO Customer VALUES  
(1,'Arun','arun@gmail.com','9876543210','Chennai'),  
(2,'Priya','priya@gmail.com','9876543211','Madurai'),  
(3,'Rahul','rahul@gmail.com','9876543212','Coimbatore'),  
(4,'Divya','divya@gmail.com','9876543213','Trichy'),  
(5,'Kavin','kavin@gmail.com','9876543214','Salem'),  
(6,'Meena','meena@gmail.com','9876543215','Chennai'),  
(7,'Vijay','vijay@gmail.com','9876543216','Erode'),  
(8,'Anu','anu@gmail.com','9876543217','Madurai'),  
(9,'Siva','siva@gmail.com','9876543218','Tirunelveli'),  
(10,'Nila','nila@gmail.com','9876543219','Thanjavur');
```

Insert Seller Data

```
INSERT INTO Seller VALUES  
(1,'Tech Store','tech@gmail.com','90000000001'),  
(2,'Fashion Hub','fashion@gmail.com','90000000002'),  
(3,'Home Store','home@gmail.com','90000000003'),  
(4,'Mobile World','mobile@gmail.com','90000000004'),  
(5,'Book House','book@gmail.com','90000000005');
```

Insert Product Data

```
INSERT INTO Product VALUES  
(1,'Laptop',55000,10,1),
```

(2,'Smartphone',25000,20,4),
(3,'Headphones',2000,30,1),
(4,'T-Shirt',800,50,2),
(5,'Shoes',1500,25,2),
(6,'Table Lamp',1200,15,3),
(7,'Keyboard',1500,20,1),
(8,'Notebook',100,100,5),
(9,'Smart Watch',4000,15,4),
(10,'Backpack',1800,20,2);

Insert Orders Data

INSERT INTO Orders VALUES

(1,1,1,1,'2026-09-01','Delivered'),
(2,2,2,1,'2026-09-02','Shipped'),
(3,3,3,2,'2026-09-03','Delivered'),
(4,4,4,3,'2026-09-04','Pending'),
(5,5,5,1,'2026-09-05','Delivered'),
(6,6,6,1,'2026-09-06','Shipped'),
(7,7,7,1,'2026-09-07','Pending'),
(8,8,8,5,'2026-09-08','Delivered'),
(9,9,9,1,'2026-09-09','Shipped'),
(10,10,10,2,'2026-09-10','Pending');

Insert Payment Data

INSERT INTO Payment VALUES

(1,1,'2026-09-01',55000,'UPI','Paid'),
(2,2,'2026-09-02',25000,'Card','Paid'),
(3,3,'2026-09-03',4000,'UPI','Paid'),
(4,4,'2026-09-04',2400,'Cash','Pending'),
(5,5,'2026-09-05',1500,'Card','Paid'),

```
(6,6,'2026-09-06',1200,'UPI','Paid'),  
(7,7,'2026-09-07',1500,'Card','Pending'),  
(8,8,'2026-09-08',500,'UPI','Paid'),  
(9,9,'2026-09-09',4000,'Card','Paid'),  
(10,10,'2026-09-10',3600,'UPI','Pending');
```

SELECT Queries

```
SELECT * FROM Customer;
```

```
SELECT * FROM Seller;
```

```
SELECT * FROM Product;
```

```
SELECT * FROM Orders;
```

```
SELECT * FROM Payment;
```

```
SELECT with Condition
```

```
SELECT * FROM Product
```

```
WHERE Price > 2000;
```

```
SELECT * FROM Orders
```

```
WHERE Status = 'Delivered';
```

UPDATE Queries

```
UPDATE Product
```

```
SET Price = 57000
```

```
WHERE Product_ID = 1;
```

```
UPDATE Orders
```

```
SET Status = 'Delivered'
```

```
WHERE Order_ID = 4;
```

DELETE Queries

-- First delete payment and order because of foreign key

DELETE FROM Payment

WHERE Payment_ID = 10;

DELETE FROM Orders

WHERE Order_ID = 10;

DELETE FROM Product

WHERE Product_ID = 10;

Create View

CREATE VIEW Customer_Order_View AS

SELECT Customer.Name, Orders.Order_ID, Orders.Status

FROM Customer

JOIN Orders

ON Customer.Customer_ID = Orders.Customer_ID;

Display View

SELECT * FROM Customer_Order_View;

JOIN Query

SELECT

c.Name,

p.Product_Name,

o.Quantity,

o.Status

FROM Customer c

JOIN Orders o

ON c.Customer_ID = o.Customer_ID

JOIN Product p

ON o.Product_ID = p.Product_ID;

Output Screenshots

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
Name	Product_Name	Quantity	Status				
Arun	Laptop	1	Delivered				
Priya	Smartphone	1	Shipped				
Rahul	Headphones	2	Delivered				
Divya	T-Shirt	3	Delivered				
Kavin	Shoes	1	Delivered				
Meena	Table Lamp	1	Shipped				
Vijay	Keyboard	1	Pending				
Anu	Notebook	5	Delivered				
Siva	Smart Watch	1	Shipped				

Customer 1 Seller 2 Product 3 Orders 4 Payment 5 Product 6 Orders 7 Customer_Order_View 8 Result 9 x

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
Name	Order_ID	Status				
Arun	1	Delivered				
Priya	2	Shipped				
Rahul	3	Delivered				
Divya	4	Delivered				
Kavin	5	Delivered				
Meena	6	Shipped				
Vijay	7	Pending				
Anu	8	Delivered				
Siva	9	Shipped				

Customer 1 Seller 2 Product 3 Orders 4 Payment 5 Product 6 Orders 7 Customer_Order_View 8 x Result 9

Output :

Result Grid							Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
Order_ID	Customer_ID	Product_ID	Quantity	Order_Date	Status					
1	1	1	1	2026-09-01	Delivered					
3	3	3	2	2026-09-03	Delivered					
5	5	5	1	2026-09-05	Delivered					
8	8	8	5	2026-09-08	Delivered					
NULL	NULL	NULL	NULL	NULL	NULL					

Customer 1 Seller 2 Product 3 Orders 4 Payment 5 Product 6 Orders 7 x Customer_Order_View 8 Result 9

Result Grid					
Filter Rows:		Edit:			
Product_ID	Product_Name	Price	Stock	Seller_ID	
1	Laptop	55000.00	10	1	
2	Smartphone	25000.00	20	4	
9	Smart Watch	4000.00	15	4	
NULL	NULL	NULL	NULL	NULL	

Customer 1	Seller 2	Product 3	Orders 4	Payment 5	Product 6	Orders 7	Customer_Order_View 8	Result 9
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Result Grid						
Filter Rows:		Edit:				
Payment_ID	Order_ID	Payment_Date	Amount	Payment_Method	Payment_Status	
1	1	2026-09-01	55000.00	UPI	Paid	
2	2	2026-09-02	25000.00	Card	Paid	
3	3	2026-09-03	4000.00	UPI	Paid	
4	4	2026-09-04	2400.00	Cash	Pending	
5	5	2026-09-05	1500.00	Card	Paid	
6	6	2026-09-06	1200.00	UPI	Paid	
7	7	2026-09-07	1500.00	Card	Pending	
8	8	2026-09-08	500.00	UPI	Paid	
9	9	2026-09-09	4000.00	Card	Paid	
10	10	2026-09-10	2500.00	UPI	Pending	

Customer 1	Seller 2	Product 3	Orders 4	Payment 5	Product 6	Orders 7	Customer_Order_View 8	Result 9
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Result Grid					
Filter Rows:		Edit:			
Product_ID	Product_Name	Price	Stock	Seller_ID	
1	Laptop	55000.00	10	1	
2	Smartphone	25000.00	20	4	
3	Headphones	2000.00	30	1	
4	T-Shirt	800.00	50	2	
5	Shoes	1500.00	25	2	
6	Table Lamp	1200.00	30	3	
7	Keyboard	1500.00	20	1	
8	Notebook	100.00	100	5	
9	Smart Watch	4000.00	15	4	
10	Backpack	1800.00	20	2	

Customer 1	Seller 2	Product 3	Orders 4	Payment 5	Product 6	Orders 7	Customer_Order_View 8	Result 9
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Limit to 1000 rows

```
1 • DROP DATABASE IF EXISTS ecommerce;
2 CREATE DATABASE ecommerce;
3 • USE ecommerce;
4 • CREATE TABLE Customer (
5     Customer_ID INT PRIMARY KEY,
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	Customer_ID	Name	Email	Phone	Address
▶	1	Arun	arun@gmail.com	9876543210	Chennai
	2	Priya	priya@gmail.com	9876543211	Madurai
	3	Rahul	rahul@gmail.com	9876543212	Coimbatore
	4	Divya	divya@gmail.com	9876543213	Trichy
	5	Kavin	kavin@gmail.com	9876543214	Salem
	6	Meena	meena@gmail.com	9876543215	Chennai
	7	Vijay	vijay@gmail.com	9876543216	Erode
	8	Anu	anu@gmail.com	9876543217	Madurai
	9	Siva	siva@gmail.com	9876543218	Tirunelveli
	10	...	...	...	...

Customer 1 x Seller 2 Product 3 Orders 4 Payment 5 Product 6 Orders 7 Customer_Order_View 8 Result 9 Apply

```
1 • DROP DATABASE IF EXISTS ecommerce;
2 CREATE DATABASE ecommerce;
3 • USE ecommerce;
4 • CREATE TABLE Customer (
5     Customer_ID INT PRIMARY KEY,
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	Seller_ID	Seller_Name	Email	Phone
▶	1	Tech Store	tech@gmail.com	9000000001
	2	Fashion Hub	fashion@gmail.com	9000000002
	3	Home Store	home@gmail.com	9000000003
	4	Mobile World	mobile@gmail.com	9000000004
	5	Book House	book@gmail.com	9000000005
*	NULL	NULL	NULL	NULL